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CS 472: Introduction to Deep Learning

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Project 1: Credit Risk Classification Using Loan Application Data

Abstract

In this project, I worked with loan application and credit history data from previous borrowers in order to design a supervised learning system to assess credit risk. This predictive model determined whether a loan applicant was more likely to default on a loan or successfully repay it, based on various financial, demographic, and behavioral features. Throughout this project, I preprocessed structured data, identified relevant features, trained classification models, evaluated their performance and classified future loan applications as either high-risk or low-risk. Ultimately, this project highlights the significant role credit risk plays in banking and financial institutions when determining loan approval decisions, financial losses, and risk management strategies.

Introduction

Banks or other financial institutions face significant losses when borrowers fail to repay their loans. The goal of this project is to create a model that predicts which borrowers would repay or default on payments. These predictions will be based on their financial and demographic attributes. Based on these attributes, two machine learning models, Logistic Regression and Random Forest, will be developed and evaluated. Additionally, a train-test split

of 80% training, 20% testing will be implemented in order for the model to identify patterns within the dataset. Thus, a hypothesis will be created in order to test the risk.

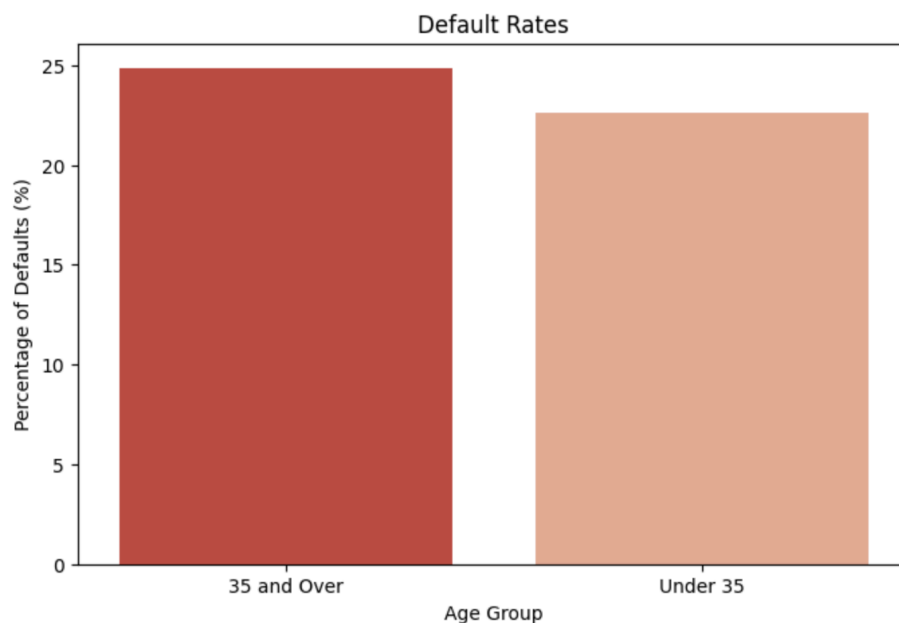
Hypothesis

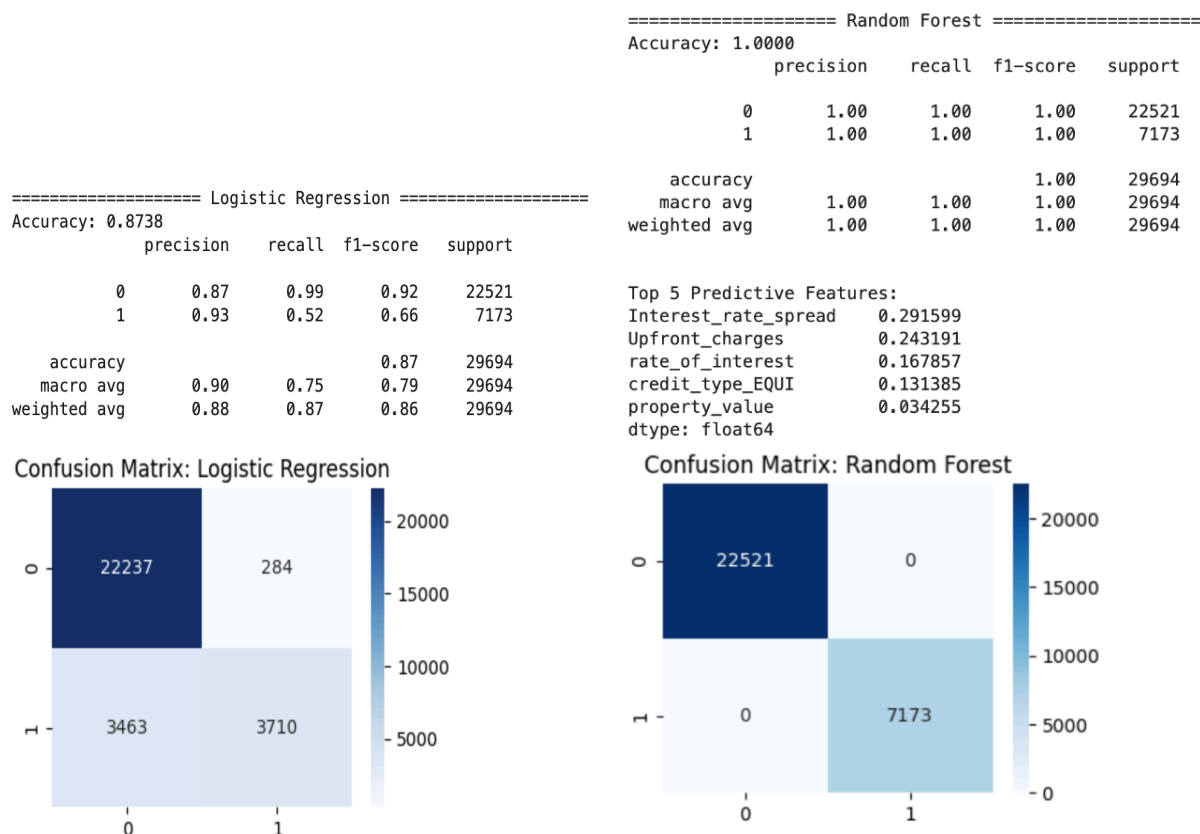
If the borrower is under the age of thirty, then they are more likely to default on payments compared to borrowers who are over the age of thirty.

Methods

For this project I gathered my data from the online website, Kaggle. This dataset, called Loan Default Dataset, was created by M Yasser H. The dataset contains past data from previous loan borrowers and the multiple deterministic factors that contribute to repayment or defaulting. The dataset consists of 148,670 rows and 34 columns. For the purpose of this project, I will focus on age, specifically the borrowers under the age of thirty and borrowers over the age of thirty.

Results





Conclusion

Based on my results, my initial hypothesis failed. The data revealed that borrowers in the 35 and Over category exhibited a higher default rate than borrowers Under 35. This suggests that age is not a clear predictor to prove which age group is riskier for financial institutions to determine which applicants would more likely default or repay. A possible reason for older borrowers being more likely to default is that they may face more financial pressures such as family expenses, higher debt to income ratio or mortgage payments can contribute to a higher default frequency. The two models that were implemented to solve the problem statement were Logistic Regression and Random Forrest. For the Logistic Regression model, it received a score of 87% and it serves as a baseline, but it shows a major weakness in catching risky borrowers. For example, the model was only able to recall 52% of defaults. Of course, this means that it

failed to identify nearly half of the high-risk borrowers. On the other hand, the Random Forrest model was a better performer with a perfect score (100%) and it was significantly more effective at identifying complex patterns. The most significant predictors that contributed to this score were Interest_rate_spread and Upfront_speed. While this model shows that it is highly capable of identifying patterns, this high score suggests that the financial features are extremely strong indicators to predict burrow status. Of course, this is good for banks because the Top 5 Predictive Features indicate that they are the most effective way to catch a default before it happens. In conclusion, even though my initial hypothesis was disproven, the project successfully delivered a Random Forest model with a 100%, which provides financial institutions with a powerful tool to detect risk by focusing on high impact indicators.

Work Cited

M Yasser H. "Loan Default Dataset." *Kaggle*,

<https://www.kaggle.com/datasets/yasserh/loan-default-dataset/data>. Accessed 12 March 2026.